

Folding Angle Control of Inter-Connected Pouch Motors

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Figure 1: Overall workflow. We first prepare (a) a 3D model and (b) generate 2D patterns. Then, we (c) laser-weld pouch motors and attach them to the panels. Lastly, we (d) actuate pouch motors by syringe, which results in an earth.

ABSTRACT

Recently, planar pneumatic actuators, known as pouch motors, have gained attention in the fields of soft robotics and Human-Computer Interaction (HCI). However, achieving complex 3D shapes by pouch motors has been hindered by two challenges: (1) the slow operating speed of customized CNC machines, conventionally used for pouch fabrication and (2) the lack of control over the folding angle of individual pouches. To overcome these limitations, (1') we introduce the use of a laser cutter to fabricate pouch motors. This approach enhances the fabrication process, enabling faster production of the pouch motors. Additionally, (2') we propose laser-welding patterns that incorporate small barriers to control the folding angle of each pouch. By manipulating the welding patterns, we achieve nearly linear control over the folding angle of the pouches. In this paper, we present the design and fabrication process of laser-welded planar pneumatic actuators and showcase several design examples.

CCS CONCEPTS

• **Applied computing** → **Computer-aided design**; *Computer-aided manufacturing*; • **Computing methodologies** → *exploration algorithms*.

KEYWORDS

planar pneumatic actuators, digital fabrication, laser cutter

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1 INTRODUCTION

In recent years, there has been significant research in the field of soft robotics focusing on developing thin, air-driven actuators known as Pouch Motors [5, 6]. In the field of Human-Computer Interaction (HCI), these actuators have found applications in shape-changing interfaces driven by liquid or gas [2, 7, 8]. However, traditional pouch motors have faced challenges when it comes to achieving complex structures. They have been limited to simple 3D shapes, which makes it difficult to create intricate designs. Two main factors contribute to these limitations. First, the conventional heat sealing method of fabricating free-form pouches with customized CNC machines [5, 6] is slow when fabricating complex shapes. Furthermore, these customized CNC machines can be challenging for regular users to operate because these machines are uncommon. Second, controlling the folding angle of individual pouches when actuated by a single input has been impossible. With the conventional pouch motors, the number of hinges required increases as the target 3D origami structure gets complex, leading to a higher number of inputs needed for actuation.

To address these limitations, we have developed novel solutions. First, we have introduced the use of a laser cutter for pouch motor fabrication, enabling rapid production at a significantly increased speed of 86 mm/sec (approximately 17 times faster than existing methods [5]). Second, by incorporating small barriers into the welding patterns, we have achieved nearly linear control over the folding angle ($< 90^\circ$) of fully inflated pouches.

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In this paper, we present the fundamental design and fabrication process of laser-welded planar pneumatic actuators, along with various design examples.

2 RELATED WORK

Numerous studies have delved into planar pneumatic actuators. One example is Pouch Motors [6], thin planar pneumatic actuators, which are fabricated by attaching two films together using a heat sealer or a soldering iron and actuated by an air pump.

There are several studies built upon the work of Pouch Motors [6]. For instance, Sticky Actuators [7] leveraged planar actuators with a sticky back to induce motion in everyday objects. PneuUI [12] explored the potential of pneumatically controlled actuators as interfaces. Liquid Pouch Motors [5] inflated pouch motors via a liquid-to-gas phase change reaction.

While these studies achieved 3D shapes by inflation, the motion was typically limited to a fixed angle of approx. 95.3° when fully inflated [6]. Although air pressure can control a single pouch motor's folding angle, simultaneously controlling multiple pouches with different angles is impossible. To overcome this limitation, we propose the use of laser-welding patterns instead of air pressure to control the folding angle.

Furthermore, these studies often relied on custom-made machines (by combining a soldering iron and CNC machine), making them less accessible to users. Moreover, the sealing speed of these custom-made machines was relatively slow, around 5.0 mm/s [5], resulting in significant time consumption for fabricating complex origami objects. To address this limitation, our study suggests the use of a laser cutter, because it is a readily accessible tool that offers rapid fabrication capabilities.

Aside from research directly related to Pouch Motors [6], there are other investigations focusing on planar pneumatic actuators. AeroMorph [8] examined the use of inflatables to generate transformable shapes through bending motion. MilliMorph [2] also achieved transformable shapes through bending motion but at a smaller scale. However, these studies mainly achieved bending rather than folding, limiting their applicability to origami structures.

Regardless of planar pneumatic actuators, researchers in HCI have explored different pneumatic interfaces and fabrication methods. For example, InflatableMod [4] combined electrohydrodynamic pump and Liquid Pouch Motors [5] for reconfigurable and pneumatic swarm robots. Sustainflatable [3] converted ambient energy into compressed air for storage, actuating it as per the user's needs. Our techniques add to such examples another way of customizing complex pneumatic structures in an easier way.

It is worth noting that while Accordion Fab [11] investigated laser-welding of thermoplastic films, it mainly focused on creating balloon-like structures using basic welding patterns like circles and squares. Consequently, the precise control of folding angles for intricate laser-welding patterns to attain 3D origami structures remains unrealized. Additionally, Panetta et al. [9] focused on fabricating curved surfaces using pouch motors, but primarily relied on linear deformations rather than rotational deformations, limiting its ability to achieve free-form origami structures.

3 LASER-WELDING PROCESS

3.1 System Details

Laser-welding pouch motors requires careful calibration and stabilization of the fabrication process. To address this, we narrow down the details. Our system uses a thermoplastic film with a thickness of 100 μm . The fabrication process involves a Trotec Speedy400 laser cutter. An add-on attached to the laser cutter's nozzle as shown in Figure 1c and Figure 5c mitigates the issue of films floating in mid-air during the operation. To guarantee comprehensive laser-welding, we repeat the procedure three times. To decide further detail, we explored several parameters in Section 3.2.

3.2 Parameter Exploration

Materials. Our study found that single-layered polyethylene sheets offered significantly better performance in the laser-welding process compared to multi-layered LLDPE (Linear Low Density Polyethylene) and ONy (Oriented Nylon) films. To compare these materials, we conducted laser-welding tests under certain conditions: Power was set to 50% (40.0 W), Speed was 2.0% (86 mm/s), and PPI was 1000. The use of two-layer films resulted in a rough, ditch-like texture (Figure 2a). The photos of the two-layer film taken by a microscope (100x) show that the welded part displays many bubbles, which could lead to air leakage. On the other hand, single-layered films formed a smooth and well-defined weld (Figure 2b). The photos taken by a microscope (100x) showed almost no bubbles in the welded part (the right half of the microscopic picture). Given these properties and the film's durability, we chose single-layered film for pouch fabrication.

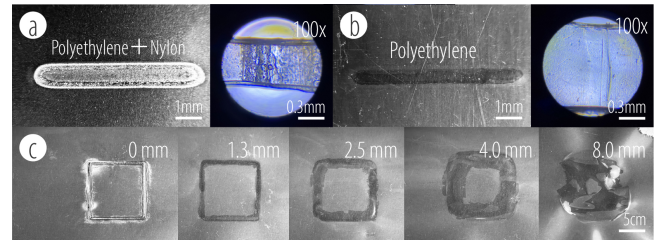


Figure 2: (a,b) Material exploration. (c) Focal point exploration.

Focal Point. Unlike conventional processes that require focused beams, our application demanded defocusing to avoid concentrated heat. Thus, we intentionally offset the laser focus from the film. This offset was found through detailed experimentation (Figure 2c). Full focus (0 cm) led to the sheets being cut due to excessive power concentration, while larger offsets (>2.5 cm) produced broadened line widths, leading to instability. Consequently, we determined an offset of 1.3 cm to be appropriate for our purposes.

Laser Settings. It was observed that a power setting above 50% ensured stability in the laser cutter. Lower power settings forced the laser cutter to exert excessive effort in suppressing power, leading

to unstable operation. With the laser output adjustments made, successful laser-welding was achieved using the following settings: Power: 50% (=40.0 W), Speed: 2.0% (=86 mm/s), and PPI: 1000.

Temperature. To maintain consistent laser cutting performance, we ensured a stable room temperature of 25 °C. In addition, we warmed up the laser cutter for at least one hour before its intended use. This warm-up period helped stabilize the laser cutter's power output, leading to more reliable and consistent performance during the fabrication process.

4 WORKING PRINCIPLE

4.1 Layer Structure

Pouch motors for folding motion typically involve two components: panels that become facets of 3D shapes and pouch motors that control the angle between adjacent panels (Figure 3a). Folding is achieved by inflating the pouch located at the boundary of two panels with air, causing the panels to lift up (Figure 3b). To achieve angles greater than 90°, a double-hinge technique was employed by aligning two pouches as illustrated in Figure 3c.

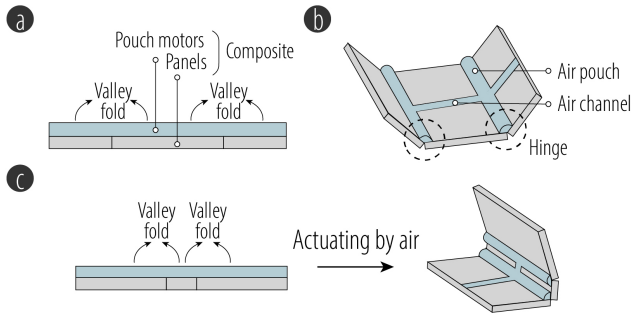


Figure 3: A schematic image of the layer structure. A composite structure of pouch motors and panels (a) and its folding motion (b). (c) The folding motion of double hinges with over 90°.

4.2 Folding Angle Control

We propose the incorporation of barriers in a zigzag arrangement (Figure 4a) within the pouch to control the folding angle upon full inflation, which was not done in previous studies based on Pouch Motors [5–7]. The unique structure of the pouch, resulting from this technique, is further elucidated through the next two experiments.

We first determine the suitable aspect ratio of the air pouch (Figure 4a). Previous research on Pouch Motors [6] has shown that a fully inflated air pouch without barriers folds at around 95.3°. To analyze the suitable aspect ratio, we conducted an experiment varying the aspect ratio of air pouches without barriers in increments of 1:1, 2:1, 3:1, 4:1, 5:1, 8:1, 10:1 and 20:1, creating a total of 24 units for each variation. The result is presented in Figure 4b. Our findings suggest that air pouches with aspect ratios between 4:1 and 20:1 fold approximately at 90°, whereas an air pouch with an aspect ratio of 1:1 notably folds close to 50°. Therefore, aspect ratios exceeding 4:1 closely approximate the desired performance

for optimal folding angles. Thus, an aspect ratio we determined on 8:1. We note that this result corresponds with the prior work on the suitable aspect ratio of the pouch motors [10].

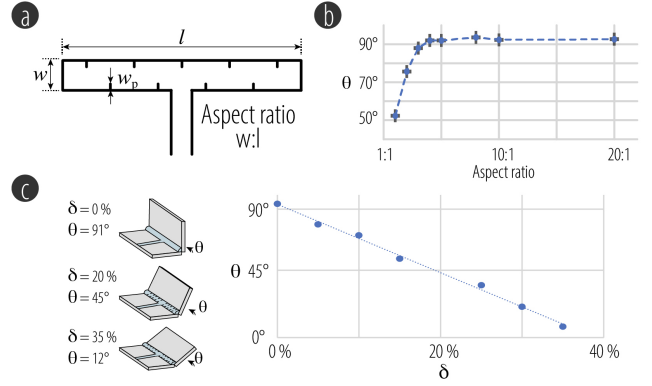


Figure 4: (a) Basic pouch structure. (b) Experiments on different aspect ratio. (c) Relationship between barrier ratio and folding angle.

To manage the folding angle θ , we introduced a barrier into the rectangular pouch design to prevent air inflow, aiming to achieve a folding angle of less than 90°. We propose zigzag placement as illustrated in Figure 4a.

Each rectangular pouch measured 20 mm × 160 mm (i.e., the aspect ratio of 8:1). The experiments involved multiple lengths of the barriers (w_p) relative to the pouch width (w). This ratio of the barrier length to the pouch width is denoted as δ , as per the following equation:

$$\delta = \frac{w_p}{w} \quad (1)$$

We examined δ at increments of 0%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, and 50% of the air pouch width. An air pressure of 150 kPa was applied to the pouch to observe the resulting folding angle. The relationship of the aligned placement was approximated by the linear equation:

$$\theta = -161\delta + 101 \quad (2)$$

By leveraging this linear relationship, we can calculate the necessary barrier lengths to attain target 3D shapes.

5 DESIGN EXAMPLES

The target 3D structure is prepared (Figure 5a) and developed into a 2D format using the Rhinoceros CAD software. The folding angles of each hinge are measured from the target 3D structure. Then, using Equation 2, the barrier patterns are prepared and incorporated into the pouch motors (Figure 5b). The subsequent step involves laser-welding the interconnected pouch motors. To aid in the fabrication, attachments are used to secure the films and prevent them from floating (Figure 5c). The resulting fabricated pouch motors are shown in Figure 5d. Finally, by applying air pressure by a syringe (Figure 5e) from a single input (Figure 5f), we achieve the desired 3D shapes.

Lastly, we fabricated various polyhedra including a hexahedron, an octahedron, and an icosahedron (Figure 6). As a design example,

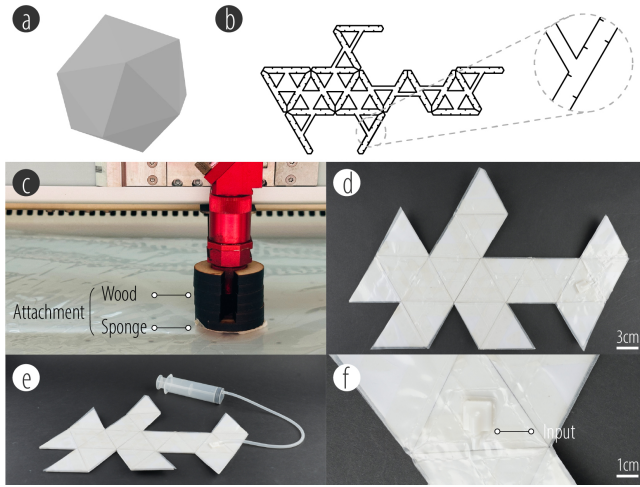


Figure 5: Overview of the research. (a) 3D model. (b) 2D laser-welding patterns. (c) Laser-welding polyethylene film by laser cutter. (d) Fabricated pouch motor. (e) Air inflation by syringe. (f) Air input.

we also designed a foldable earth by pasting the map visual (gained from [1] under permission) to the surface of the pouch composite.

	2D pattern		3D model	3D shape
a		Name: octahedron Single hinge: 7 Double hinge: 0 Folding angle: 71		
b		Name: icosahedron Single hinge: 3 Double hinge: 2 Folding angle: 39 110		
c		Name: icosahedron Single hinge: 19 Double hinge: 0 Folding angle: 42		

Figure 6: Demonstrations of several design examples.

6 CONCLUSION

In this paper, we presented a technique for achieving free-form 3D folding shapes in pouch motors by applying air pressure. In order to facilitate the manufacturing process of pouch motors, we plan to tackle the inverse problem by developing software that automatically converts user-input 3D mesh structures into 2D patterns of pouch motors and panels. This software would greatly simplify the production of pouch motors by eliminating the need for manual pattern generation. By addressing this challenge, we anticipate enhanced efficiency and broader application of pouch motors as origami structures.

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